

The Knee

- Primarily functions as a hinge joint with the articulation b/t the femur and the tibia serving as the main load bearing
- Distal femur → bone widens to form Medial and lateral condyles
- Distal femur and proximal tibial articular surfaces are covered with hyaline cartilage
- Patella → large sesamoid bone that functions as the insertion of the quadriceps tendon and the origin of the infrapatellar ligament
 - Anterior stabilization and
- Fibula → Allow rotation of the lower leg without sacrificing stability
 - The proximal end of the fibular head articulates medially with the lateral portion of the tibial plateau

The Knee - Ligaments

- Anterior Cruciate Ligament (ACL)
 - Crosses anterior medial tibia to the lateral femoral condyle (prevents tibia from sliding forward on the femur)
- Posterior Cruciate Ligament (PCL)
 - Crosses posterior tibia to the medial femoral condyle (prevents tibia from sliding backward on the femur)
- Medial Collateral Ligament (MCL)
 - Connects medial femoral epicondyle to the medial condyle of the tibia
 - Also attaches to the medial meniscus
- Lateral Collateral Ligament (LCL)
 - Connects lateral femoral epicondyle and the head of the fibula

Hip Exam- Palpation

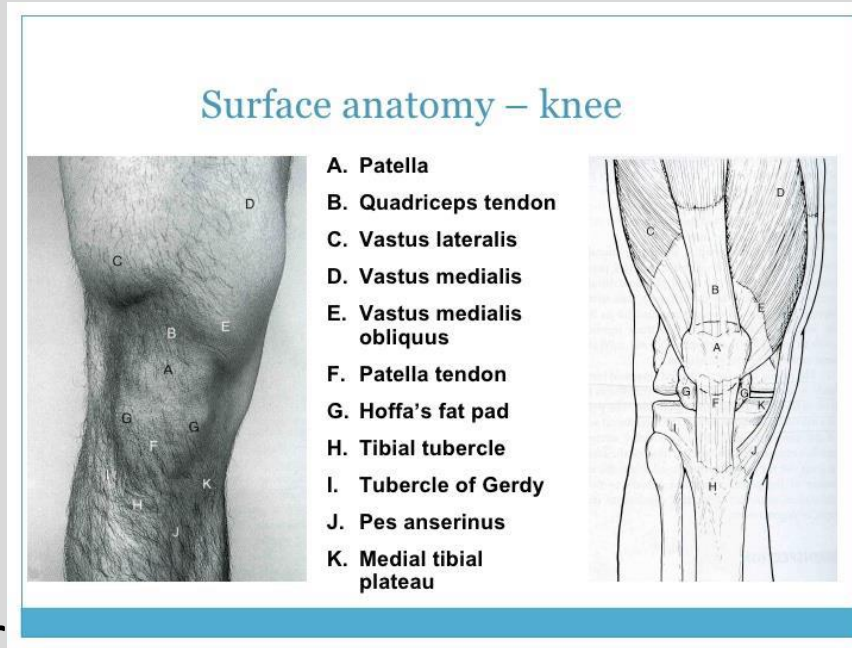
- Bilateral exam
- Use finger pads for palpation
- Patient in lateral recumbent position
- Inguinal region – check for lymph nodes or hernia
- Note: Ligament or tendon tenderness after injury is suspicious for tear



- Landmarks
 - Greater Trochanter
 - Anterior Superior Iliac spine (ASIS)
 - Anterior Inferior Iliac spine (AIIS)
 - Posterior superior iliac spine (PSIS)
 - Sacroiliac joint
 - Pubic tubercle
 - Iliac crest
 - Ischial tuberosity

Knee Exam- Palpation

- Bilateral palpation
- Use finger pads for palpation
- Patient in seated position
- Palpate for tenderness



- Landmarks
 1. Patella
 2. Tibial tubercle
 3. Medial / lateral epicondyle of femur
 4. Medial / lateral epicondyle of tibia
 5. Medial / lateral knee joint line
 6. Medial / lateral collateral ligament.
 7. Patellar ligament
 8. Iliotibial band
 9. Fibular head at its styloid process

Hip Exam- Passive Range of Motion Testing

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- Bilateral exam
- Gown covering patient's thighs and above



Flexion



Extension



External Rotation



ABDuction



ADDuction



Internal Rotation

Source: Bates' Guide to
the Physical Examination
and History Taking; 12th
Ed. – Chapter 16

Hip Exam- Active Range of Motion Testing

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- Bilateral exam
- Gown covering patient's thighs and above



Flexion



Extension



External Rotation



ABDuction



ADDuction



Internal Rotation

Source: Bates' Guide to
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Knee Evaluation – Range of Motion

Knee Movement	Primary Muscles Affecting Movement	Patient Instructions
Flexion	Hamstring group: biceps femoris, semitendinosus, and semimembranosus	<i>“Bend or flex your knee.” OR “Squat down to the floor.”</i>
Extension	Quadriceps: rectus femoris, vastus medialis, lateralis, and intermedius	<i>“Straighten your leg.” OR “After you squat down to the floor, stand up.”</i>
Internal Rotation	Sartorius, gracilis, semitendinosus, semimembranosus	<i>“While sitting, swing your lower leg toward the midline.”</i>
External Rotation	Biceps femoris	<i>“While sitting, swing your lower leg away from the midline.”</i>

Trendelenburg's Test

- The Patient stands first on one leg on the unaffected side, then the affected side, while observing for a continued pelvis
- *Positive Test:* The pelvis on the opposite side drops
- *Indicates:* Inability of the hip abductors to stabilize the pelvis on the femur indicates a weak or painful gluteus medius on the weight-bearing leg

Source: ICC Guide

Video Credit: NYITCOM Bears and Red Wolves

Trendelenburg's Test

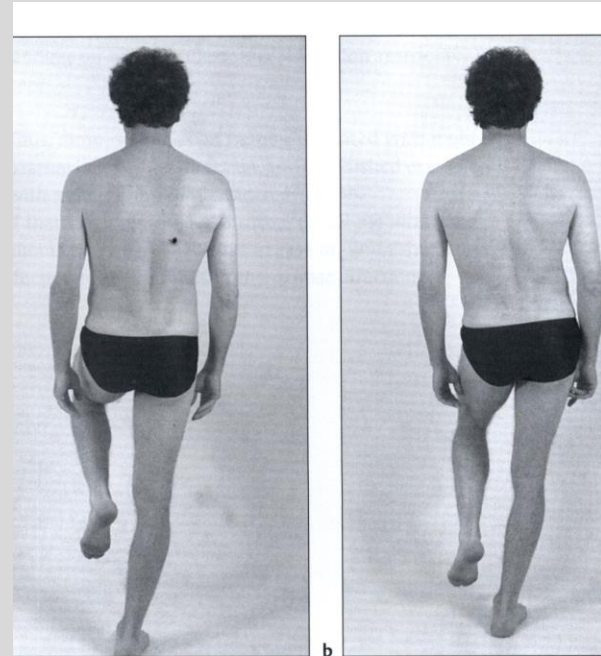
- Abnormal: Iliac crest drops on contralateral side of weakness (gluteus medius weak on standing leg)

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Normal

Abnormal



Stinchfield Test

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- The patient is supine with knee in extension. The patient actively raises leg off table while examiner applies manual resistance.
- Positive test: Reproducible pain usually in the groin, gluteal region and greater trochanter region
- Indicates: Intra-articular hip pathology

Video Credit: NYITCOM Bears
and Red Wolves

Patrick's (FABER) Test

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- Indicates the motions engaged:
 - Flexion
 - ABduction
 - External Rotation
 - Extension
- The examiner then pushes the knee toward the table to further stress the hip and SI joint. If needed, the 2nd hand can be placed on the opposite ASIS and can press down simultaneously on both sides.
- Positive test: There is pain from these movements or if patient is unable to perform them
- Indicates: Hip joint pathology, SI dysfunction, muscular problem depending on location of the pain

Log Roll

- Nonspecific exam of the hip joint
- Patient supine
- Position the affected leg on the side of the examiner
- The knee is held in extension by the examiner and the foot is internally and externally rotated.
- Positive Test: Pain with motion. A negative evaluation does not preclude hip pathology
- Indicates: Non-specific or more ominously could represent fracture of the hip
- “Popping” sound may be due to acetabular labral tear

FADIR / FAIR Test

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- Indicates motions engaged: Flexion, Adduction, Internal Rotation
- Look for quantity , quality , and symmetry of motion
- Could indicate problem with hip joint (Impingement) or muscular problem (Piriformis)
- Positive test when there is pain from these movements or if patient is unable to perform them

Video Credit: NYITCOM Bears
and Red Wolves

Note: Slide reproduced from PowerPoint Presented to
Class of 2022- *Foundations of Osteopathic Medicine*-
OMM Lab #11 – “Osteopathic Evaluation of the Hip”
presented November 14th and 15th, 2018

Thomas Test

- Patient supine.
 - Flexes both knees to chest.
 - Allows one leg to extend.
 - Measure the distance (of extended leg) between popliteal crease to the table
- Positive test: The patient is unable to bring leg down to full extension or if there are compensatory changes in the lumbar spine with full extension
- Indicates: Hip flexor (usually psoas muscle) hypertonicity

Normal

Abnormal

Erichsen's Test

- Patient supine
 - Examiner places hands on ilium bilaterally
 - Applies medial pressure towards sacroiliac joints bilaterally
- Positive test: Pain at the SI joints
- Indicates: Sacroiliitis at the side of pain

Ober's Test

- Patient lying on side with the affected side up.
 - The examiner is behind the patient
 - Hip and knees are flexed
 - The upper leg is passively abducted and extended.
 - The limb is then lowered to the table
 - The hip is stabilized to prohibit the patient from rolling backward and allowing the ITB to slip over the greater trochanter
- Positive test: The leg will not fall to the table or there is a delay in drop
- Indicates: Iliotibial Band (ITB) Contracture

Normal

Abnormal

Sources:

1. Stockard, Alan, Osteopathic Clinical Joint Exam; American Osteopathic Academy of Sports Medicine; Chapter 7
2. Note: Slide reproduced from PowerPoint Presented to Class of 2022- *Foundations of Osteopathic Medicine*- OMM Lab #11 – “Osteopathic Evaluation of the Hip” presented November 14th and 15th, 2018

Video Credit: NYITCOM Bears and Red Wolves

Piriformis Test

- Patient supine
 - Flexed hip and knee
 - Active ABduction against examiner resistance
- Positive test: Pain over posterior aspect of the greater trochanter
- Indicates: Piriformis spasm or pathology

Hip Drop Test (Lateral Lumbar Flexion Test)

- The side of the greatest drop of the iliac crest, is opposite to the side of the side of the greatest lumbar sidebending. [Or, the lumbar convexity, is on the same side, as the side of the greater amount of drop, of the iliac crest during this test].

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Special Tests - Knee

- McMurray's Test (Meniscus)
- Apley Compression Test (Meniscus)
- Apley Distraction Test (Collateral Ligaments)
- Lachman's Test (ACL)
- Anterior Drawer Test (ACL)
- Posterior Drawer Test (PCL)
- Valgus Stress Test (MCL)
- Varus Stress Test (LCL)
- Patellar Laxity Test (Patellar Mechanism)
- Patellar Grind Test (Patellofemoral Joint Disorder)
- Thessaly Test (Meniscus)

McMurray's Test

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- Purpose is to pinch the torn meniscus between the tibial plateau and femoral condyles.
- Examiner lifts the leg of the supine patient, one hand controlling the ankle while the other controls the distal femur
- The knee is flexed
- To stress the **medial meniscus**, the tibia is internally rotated as the knee is extended adding a varus load.
- The examiner is stressing the tibia **medially** so as to compress the **medial** joint compartment, while the rotation forces the condyles on and against the **medial meniscus**
- Positive Test: Pain or a painful click while the knee is brought into extension
- Indicates: Possible meniscus tear (**Medial**)

Video Credit: NYITCOM Bears
and Red Wolves

Source: Stockard, Alan, Osteopathic Clinical
Joint Exam; American Osteopathic Academy
of Sports Medicine; Chapter 8

McMurray's Test

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- Purpose is to pinch the torn meniscus between the tibial plateau and femoral condyles.
- Examiner lifts the leg of the supine patient, one hand controlling the ankle while the other controls the distal femur
- The knee is flexed
- To stress the **lateral meniscus**, the tibia is externally rotated as the knee is extended adding a valgus load.
- The examiner is stressing the tibia **laterally** so as to compress the **lateral** joint compartment, while the rotation forces the condyles on and against the **lateral meniscus**
- Positive Test: Pain or a painful click while the knee is brought into extension
- Indicates: Possible meniscus tear (**Lateral**)

Video Credit: NYITCOM Bears
and Red Wolves

Source: Stockard, Alan, Osteopathic Clinical
Joint Exam; American Osteopathic Academy
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Apley's Compression Test

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- Patient is prone with the knee flexed to 90 degrees
- Standing at the opposite side of the of the involved leg being tested
- The clinician pushes down on the foot to provide a compression force on the meniscus while rotating the foot internally and externally
- Also, can be done with examiner grasping patient's ankle and rotating it internally and externally while a downward force is applied
- Positive Test: Pain during compression plus rotation
- Indicates: Pain on medial or lateral side may indicate meniscal damage on that side

Apley's Distraction Test

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- Patient is prone with the knee flexed to 90 degrees
- Standing at the opposite side of the of the involved leg being tested
- The clinician applies a distraction force pulling up on the foot while rotating the foot internally and externally
- Also, can be done with examiner grasping patient's ankle and rotating it internally and externally while a distraction force is applied
- Positive Test: Pain during distraction force plus rotation
- Indicates: Pain on medial or lateral side may indicate ligamentous injury on that side

Video Credit: NYITCOM Bears
and Red Wolves

Source: Stockard, Alan, Osteopathic Clinical
Joint Exam; American Osteopathic Academy
of Sports Medicine; Chapter 8

Lachman's Test

- Patient is supine
- Standing at the same side of the involved leg being tested
- Bilateral exam
- The Clinician places the proximal stabilizing hand on the distal thigh just above the patella
- The distal mobilizing hand grasps the proximal tibia
- Flex the knee to 15-30 degrees, the examiner uses his distal hand to pull the tibia anteriorly while the proximal hand stabilizes the thigh
- Positive Test: Increased laxity, soft or absent end point
- Indicates: ACL tear / insufficiency

Anterior Drawer Test

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- Standing at the same side of the involved leg being tested
- Bilateral exam
- Patient is supine position with knee flexed to 90 degrees and patient's foot is flat on the table
- The clinician sits on the patient's foot to stabilize the leg while the examiner grasps the proximal tibia with both hands and pulls the tibia anteriorly
- The examiner cups their hands around the knee with the thumbs on the medial and lateral joint lines and the fingers on the medial and lateral insertions of the hamstrings
- Positive Test: Excessive translation when compared to the other knee
- Indicates: ACL tear / insufficiency

Video Credit: NYITCOM Bears
and Red Wolves

Source: Stockard, Alan, Osteopathic Clinical
Joint Exam; American Osteopathic Academy
of Sports Medicine; Chapter 8

Posterior Drawer Test

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- Standing at the same side of the involved leg being tested
- Bilateral exam
- Patient is supine position with knee flexed to 90 degrees and patient's foot is flat on the table
- The clinician sits on the patient's foot to stabilize the leg while the examiner grasps the proximal tibia with both hands and translate the tibia posteriorly
- The examiner cups their hands around the knee with the thumbs on the medial and lateral joint lines and the fingers on the medial and lateral insertions of the hamstrings
- Positive Test: Excessive translation when compared to the other knee
- Indicates: PCL tear / insufficiency

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Source: Stockard, Alan, Osteopathic Clinical
Joint Exam; American Osteopathic Academy
of Sports Medicine; Chapter 8

Valgus Stress Test

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- Patient is supine
- Examiner stands on same side of knee being examined
- Bilateral exam
- Examiner slight flexes the patient's knee and moves the thigh to about 30 degrees laterally to the side of the table
- While stabilizing the distal femur with one hand on the lateral side of the knee and the other hand around the medial ankle, push medially against the knee and pull laterally at the ankle to open the knee joint on the medial side
- This creates a valgus load on the MCL of the knee
- Positive Test: Increased laxity, sort or absent endpoint or pain
- Indicates: Medial Collateral Ligament disruption

Video Credit: NYITCOM Bears
and Red Wolves

Source: Stockard, Alan, Osteopathic Clinical
Joint Exam; American Osteopathic Academy
of Sports Medicine; Chapter 8

Varus Stress Test

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- Patient is supine
- Examiner stands at the opposite side of knee being examined
- Bilateral exam
- Examiner slight flexes the patient's knee and moves the thigh to about 30 degrees laterally to the side of the table
- While stabilizing the knee with one hand on the medial side of the knee and the other hand around the lateral ankle, examiner push laterally against the knee and pull medially at the ankle to open the knee joint on the Lateral side
- This creates a varus load on the LCL of the knee
- Positive Test: Increased laxity, sort or absent endpoint or pain
- Indicates: Lateral Collateral Ligament disruption

Video Credit: NYITCOM Bears
and Red Wolves

Source: Stockard, Alan, Osteopathic Clinical
Joint Exam; American Osteopathic Academy
of Sports Medicine; Chapter 8

Thessaly Test

- Hold patient's outstretched hands while he or she stands flat-footed on the floor
- Internally and Externally rotated while on 1 leg, with the knee 20 degrees flexed
- Positive Test: Pain with rotational movement
- Indicates: Meniscus Injury